

Database Design Document — MoMo Analytics Platform

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Overview

The MoMo Analytics Platform database is built on MySQL and consists of six tables designed to store, categorize, tag, and audit every MoMo SMS transaction extracted from the XML data source. The schema was designed with three priorities in mind: referential integrity to prevent orphaned or inconsistent records, query performance through strategic indexing, and full traceability by preserving the original SMS text alongside every processed transaction.

Design Rationale

Why a separate `users` table? Phone numbers and names appear repeatedly across hundreds of transactions. Rather than duplicating that data in every transaction row, we extract counterparty information into a dedicated `users` table. This eliminates redundancy, enables user-level analytics (total sent, total received, activity range) with simple JOIN queries, and makes it easy to update a name without touching transaction records.

Why a separate `categories` table? The MoMo XML data contains 11 distinct transaction types identified through SMS pattern matching. Storing these as a lookup table rather than raw strings in `transactions` enforces consistency — a transaction can only belong to a category that exists — and makes it trivial to add new categories or rename existing ones without a bulk UPDATE.

Why `tags` and a junction table? Categories capture the primary transaction type (one per transaction). Tags are flexible, analyst-defined labels such as "high-value" or "flagged" that can be applied after ingestion. A single transaction can carry multiple tags, and the same tag applies to many transactions — a many-to-many relationship. The `transaction_tags` junction table resolves this cleanly with a composite primary key that also prevents duplicate tag assignments.

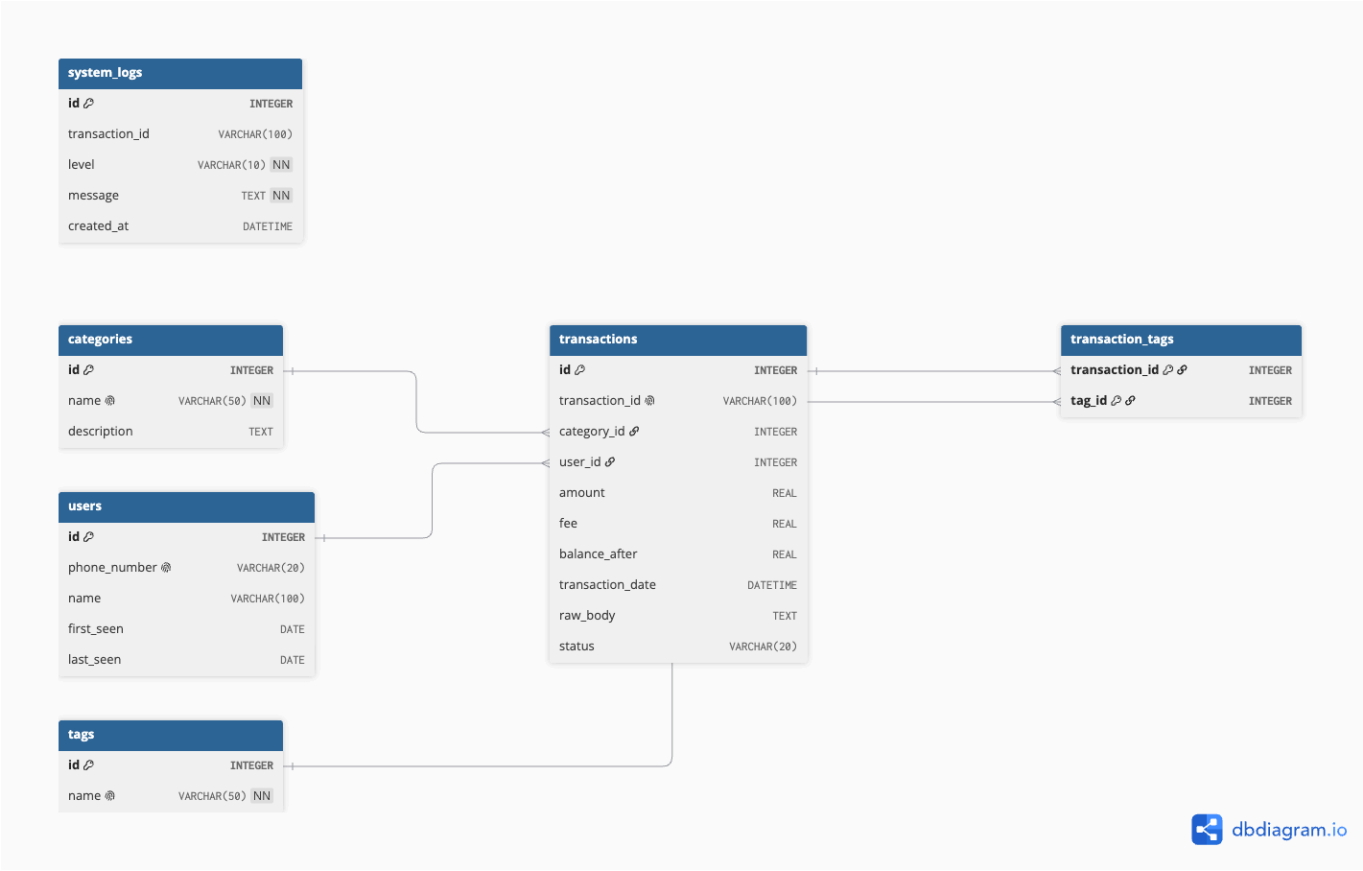
Why store `raw_body`? The original SMS text is preserved verbatim in every transaction row. This means any future improvement to the categorisation regex can be re-applied to historical data without re-importing the XML source file, and any disputed transaction can be traced back to its exact original message.

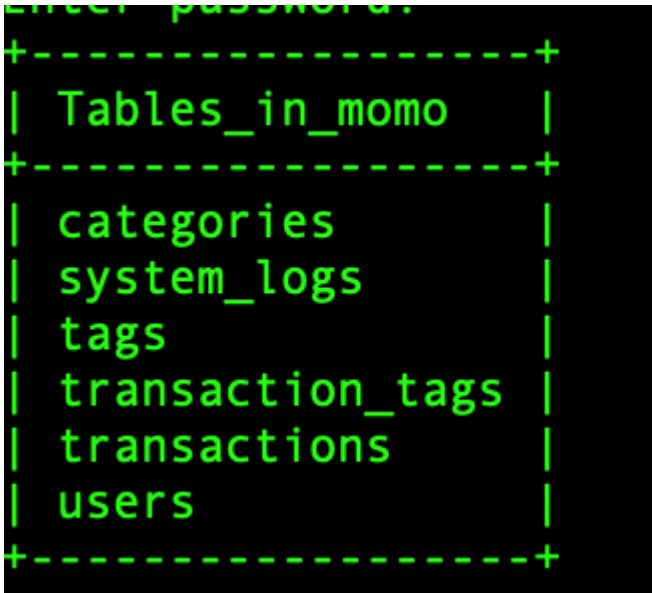
Why is `transaction_id` in `system_logs` nullable? Not every log event relates to a specific transaction. Pipeline-level events such as startup, file-not-found errors, and schema validation failures need to be recorded in the same audit trail. Making `transaction_id` nullable in `system_logs` accommodates both transaction-level and pipeline-level events in one table.

Indexes strategy: Indexes are placed on `transaction_date`, `category_id`, and `user_id` in the `transactions` table — the columns most frequently used in WHERE clauses and JOINS. Two indexes on `system_logs` cover `level` (for filtering ERROR entries) and `created_at` (for time-range queries). This avoids full-table scans on the two largest tables in the schema.

Entity Relationship Summary

Relationship	Type	Description
transactions → categories	Many-to-One (M:1)	Each transaction belongs to one category
transactions → users	Many-to-One (M:1)	Each transaction has one counterparty user
transactions ↔ tags	Many-to-Many (M:N)	Resolved via <code>transaction_tags</code> junction table
system_logs → transactions	Many-to-One (M:1)	Many log entries can reference one transaction





Data Dictionary

categories

Lookup table for the 11 MoMo transaction types. Seeded at schema creation.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Surrogate key
name	VARCHAR(50)	NOT NULL, UNIQUE	Machine-readable transaction type label
description	TEXT		Human-readable explanation of the category

users

Stores unique counterparty phone numbers and names extracted from SMS bodies.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Surrogate key
phone_number	VARCHAR(20)	UNIQUE	Counterparty phone number
name	VARCHAR(100)		Display name extracted from SMS
first_seen	DATE		Date of earliest recorded transaction
last_seen	DATE		Date of most recent transaction

transactions

Core table. One row per MoMo SMS transaction.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Surrogate key
transaction_id	VARCHAR(100)	UNIQUE	MoMo reference number from SMS
category_id	INTEGER	FK → categories.id	Transaction type
user_id	INTEGER	FK → users.id	Counterparty user
amount	REAL	CHECK (amount >= 0)	Transaction amount in RWF
fee	REAL	CHECK (fee >= 0)	Service fee in RWF
balance_after	REAL		Account balance after transaction
transaction_date	DATETIME		Timestamp parsed from SMS body
raw_body	TEXT		Original SMS text for traceability
status	VARCHAR(20)	CHECK IN ('success','failed','reversed'), DEFAULT 'success'	Transaction outcome

tags

Free-form analyst labels that can be applied to transactions after ingestion.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Surrogate key
name	VARCHAR(50)	NOT NULL, UNIQUE	Tag label e.g. high-value, flagged

transaction_tags (junction table — resolves M:N)

Links transactions to tags. A transaction can have many tags; a tag can apply to many transactions.

Column	Type	Constraints	Description
transaction_id	INTEGER	PK (composite), FK → transactions.id, ON DELETE CASCADE	
tag_id	INTEGER	PK (composite), FK → tags.id, ON DELETE CASCADE	

system_logs

Audit trail for ETL pipeline events and per-transaction processing results.

Column	Type	Constraints	Description
id	INTEGER	PK, AUTOINCREMENT	Surrogate key
transaction_id	VARCHAR(100)	nullable	Related MoMo reference (null for pipeline events)
level	VARCHAR(10)	NOT NULL, CHECK IN ('INFO','WARNING','ERROR')	Log severity level
message	TEXT	NOT NULL	Log message body
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Auto-set timestamp

Sample Queries

1. Total amount and fees per category

```
SELECT c.name,  
       COUNT(*)           AS tx_count,  
       SUM(t.amount)      AS total_amount,  
       SUM(t.fee)         AS total_fees  
FROM transactions t  
JOIN categories c ON t.category_id = c.id  
GROUP BY c.name  
ORDER BY total_amount DESC;
```

2. Top 5 users by transaction volume

```
SELECT u.phone_number,  
       u.name,  
       COUNT(*)           AS tx_count,  
       SUM(t.amount)      AS total_amount  
FROM transactions t  
JOIN users u ON t.user_id = u.id  
GROUP BY u.id  
ORDER BY total_amount DESC  
LIMIT 5;
```

3. All transactions carrying the "high-value" tag

```
SELECT t.transaction_id, t.amount, t.transaction_date, c.name AS category
FROM transactions t
JOIN transaction_tags tt ON tt.transaction_id = t.id
JOIN tags tg            ON tg.id = tt.tag_id
JOIN categories c       ON c.id = t.category_id
WHERE tg.name = 'high-value';
```

4. Error log entries from the last 7 days

```
SELECT id, transaction_id, message, created_at
FROM system_logs
WHERE level = 'ERROR'
      AND created_at >= NOW() - INTERVAL 7 DAY
ORDER BY created_at DESC;
```

5. Monthly transaction totals

```
SELECT DATE_FORMAT(transaction_date, '%Y-%m') AS month,
       COUNT(*)           AS tx_count,
       SUM(amount)        AS total_amount,
       SUM(fee)           AS total_fees
FROM transactions
GROUP BY month
ORDER BY month;
```

```
name      tx_count  total_amount  total_fees
incoming_money  1      50000.00      0.00
withdrawal_from_agent  1      30000.00      300.00
transfer_to_mobile    1      20000.00      200.00
payment_to_code_holder 1       5000.00      50.00
airtime_bill_payment  1       1000.00      0.00
phone_number  name      tx_count  total_amount
+250781000001 Alice Uwimana  1      50000.00
+250784000004 David Habimana  1      30000.00
+250782000002 Bob Nkurunziza  1      20000.00
+250785000005 Eve Ingabire   1       5000.00
+250783000003 Claire Mukamana 1       1000.00
transaction_id amount  transaction_date  category
TXN-2024-0001  50000.00  2024-08-01 09:15:00  incoming_money
TXN-2024-0004  30000.00  2024-08-15 11:45:00  withdrawal_from_agent
month tx_count  total_amount  total_fees
2024-08 5      106000.00      550.00
```

```

+-----+-----+
| transaction_id | status |
+-----+-----+
| TXN-2024-0003 | reversed |
+-----+-----+

```

```

+-----+-----+
| transaction_id | tag_id |
+-----+-----+
|              4 |      1 |
|              2 |      2 |
|              1 |      3 |
|              5 |      5 |
+-----+-----+

```

Security & Unique Constraint Rules

The following rules are enforced at the database layer, independent of application code:

1. `transactions.transaction_id` — **UNIQUE** — prevents the same SMS from being imported twice.
2. `users.phone_number` — **UNIQUE** — ensures one user record per phone number, no duplicates.
3. `categories.name` — **UNIQUE** — prevents duplicate category labels in the lookup table.
4. `tags.name` — **UNIQUE** — prevents duplicate tag labels.
5. `transaction_tags(transaction_id, tag_id)` — **composite PK** — prevents the same tag being applied to the same transaction more than once.
6. `CHECK (amount >= 0)` — rejects negative transaction amounts at insert time.
7. `CHECK (fee >= 0)` — rejects negative fee values at insert time.
8. `CHECK (status IN ('success', 'failed', 'reversed'))` — rejects any unknown status string.
9. `CHECK (level IN ('INFO', 'WARNING', 'ERROR'))` — rejects invalid log level values.
10. `ENGINE=InnoDB` — enforces all REFERENCES constraints at runtime, preventing orphaned foreign key values.

```

ERROR 1062 (23000) at line 1: Duplicate entry 'TXN-2024-0001' for key 'transactions.uq_transactions_txid'
ghz@MacBookPro codesync-week2-database-design-1 % mysql -u root -p momo -e "INSERT INTO transactions (transaction_id, amount) VALUES ('TXN-TEST', -500);"
```

```

ERROR 3019 (HY000) at line 1: Check constraint 'chk_amount' is violated.
ghz@MacBookPro codesync-week2-database-design-1 % mysql -u root -p momo -e "DELETE FROM transaction_tags WHERE transaction_id=1 AND tag_id=1; SELECT * FROM transaction_tags;"
```